THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bridge near the sea rusts faster because: 1

(a) salt in sea air speeds rusting **(b)** sea is deep
(c) sea is cold **(d)** sea is blue

2. Which of the following is a crystal? 1

(a) copper sulphate **(b)** chalk powder **(c)** cotton **(d)** flour

3. A copper vessel turns green on keeping in moist air. This is: 1

(a) a chemical change **(b)** reversible
(c) a physical change **(d)** melting

4. Which of the following does not rust? 1

(a) iron pipe **(b)** stainless steel spoon
(c) iron gate **(d)** iron nail

5. Which of these processes forms crystals of salt from sea water? 1

(a) evaporation of water **(b)** sieving
(c) filtration **(d)** magnetism

6. When a piece of chalk is crushed to powder: 1

(a) a chemical change occurs **(b)** new substance forms
(c) it burns **(d)** a physical change occurs

7. Why are iron objects like railings painted? 1

(a) to keep air and moisture away **(b)** to make them heavy
(c) to make them soft **(d)** to make them transparent

8. A student adds copper sulphate crystals to water and gets a blue solution. This is: 1

(a) rusting **(b)** a chemical change
(c) a physical change **(d)** burning

9. When a piece of chalk is dropped into dilute acid and bubbles appear: 1

(a) a chemical change occurs **(b)** nothing happens
(c) it melts **(d)** a physical change occurs

10. Why is a layer of oil or grease applied on machine parts? 1

(a) prevent contact with moisture and air **(b)** make them magnetic
(c) make them lighter **(d)** make them hot

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A piece of paper is torn and then burnt. Compare the two changes. 2

12. What is meant by a periodic change? Give an example. 2

13. Why is the dissolving of sugar in water a physical change but burning of sugar a chemical change? 2

14. Why is a banana turning black a chemical change? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain, with an example, how change of state differs from a chemical change. 3

16. Describe the steps to make crystals of salt from salt solution. 3

17. Why is the formation of dew a physical change? 3

OR

A student collected 5 g of salt, dissolved it and evaporated the water. He got 5 g of salt back. What does this prove? 3

18. Why is the melting of ice cream in a cone only a physical change? 3

OR

Describe the changes when sugar is heated. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the activity to show that a chemical change may need heat, with the example of heating sugar and heating a candle. 5

OR

Describe how to show that a gas is evolved in a chemical change and identify it. 5

20. Discuss how knowledge of chemical changes helps in food preservation. 5

OR

Explain why a student's statement 'all changes of colour are chemical' may not be true. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND